Proposal Evaluation Form

EUROPEAN COMMISSION

Horizon Europe Framework Programme (HORIZON)

Evaluation Summary Report

Call:	HORIZON-MSCA-2025-PF
Type of action:	HORIZON-TMA-MSCA-PF-EF
Proposal number:	101285973
Proposal acronym:	ScrollNets
Duration (months):	24
Proposal title:	Towards a Diagrammatic Foundation for Next-Generation Interactive Theorem Provers
Activity:	EF-ENG

N.	Proposer name	Country	Total eligible costs	%	Grant Requested	%
1	THE UNIVERSITY OF BIRMINGHAM	UK	260,347.92	100.00%	260,347.92	100.00%
	Total:		260,347.92		260,347.92	

Abstract:

Interactive Theorem Provers (ITPs) are powerful software tools for formal verification, yet their steep learning curve hinders broader adoption. The ScrollNets project addresses this challenge by developing a new foundation for ITPs based on "scroll nets", a novel diagrammatic proof formalism introduced by the researcher. Rooted in Peirce's Existential Graphs, scroll nets combine the structural flexibility of deep inference with an intuitive graphical representation, enabling proofs to be constructed via direct-manipulation gestures in a "Proof-by-Action" paradigm.

The project's primary objective is to establish scroll nets as a robust and expressive framework for a new generation of interactive, user-friendly ITPs. The research is structured into four work packages: WP1 will establish the core metatheory, including normalization and sequentialization properties. Subsequent work packages will extend the framework to richer logical systems, including classical and bi-intuitionistic logic (WP2), fixpoint logics for recursion (WP3), and dependent type theory (WP4).

By making formal reasoning more accessible, ScrollNets promises significant scientific, societal, and economic impacts. It will provide an intuitive "no-code" platform for mathematics education, empower domain experts in industrial formal verification, and enhance the reliability of neurosymbolic AI systems. The project will be hosted at the University of Birmingham's world-leading Theory of Computation group, leveraging the supervisor's expertise to ensure its success and advance the state-of-the-art in proof theory and interactive theorem proving.

Evaluation Summary Report

Evaluation Result

Total score: 91.20 % (Threshold: 70 /100.00)

Criterion 1 - Excellence

Score: **4.50** (Threshold: 0 / 5.00 , Weight: 50.00%)

- **Quality and pertinence of the project's research and innovation objectives (and the extent to which they are ambitious, and go beyond the state of the art)**
- **Soundness of the proposed methodology (including interdisciplinary approaches, consideration of the gender dimension and other diversity aspects if relevant for the research project, and the quality of open science practices)**
- **Quality of the supervision, training and of the two-way transfer of knowledge between the researcher and the host**
- **Quality and appropriateness of the researcher's professional experience, competences and skills**

Quality and pertinence of the project's research and innovation objectives:
Very good

Extent to which the research and innovation objectives are achievable, measurable and verifiable:
Good

Level of ambition of the research and innovation objectives and advancement beyond the state of the art:
Good

Soundness of the proposed methodology, including the concepts, models, and assumptions underpinning the project:
Very good

Identification of important methodological challenges and proposed measures to tackle them:
Good

Integration of interdisciplinary expertise and methods to pursue the project's objectives:
Good

Consideration of the gender dimension and other diversity aspects in the proposal’s research and innovation content:
Not applicable

Implementation of appropriate open science practices in the proposed methodology:
Excellent

Quality of the supervision:
Excellent

Quality of the planned training activities for the researcher:
Very good

Overall effectiveness of the transfer of knowledge between the researcher, the host organisation, and, if applicable, the outgoing phase host:
Excellent

Rationale and added value of the non-academic placement:
Not applicable

Rationale and added value of the secondment(s):
Not applicable

Quality of the researcher's professional experience, competences and skills:
Excellent

Appropriateness of the researcher's professional experience in relation to the proposed research project:

Excellent

Comments:

The proposal fails to properly describe the pertinence of the research and its progress beyond the state of the art.
The absence of usability testing or experimental implementation reduces practical impact and measurability.
Mitigation measures for the methodological challenges lack detail. Interdisciplinarity is conceptually strong yet not operationalised. There is no clear framework to assess the acquired skills or training deliverables.

Criterion 2 - Impact

Score: **4.50** (Threshold: 0 / 5.00 , Weight: 30.00%)

- **Credibility of the measures to enhance the career perspectives and employability of researchers and contribution to their skills development.**
- **Suitability and quality of the measures to maximise expected outcomes and impacts, as set out in the dissemination and exploitation plan, including communication activities.**
- **The magnitude and importance of the project's contribution to the expected scientific, societal and economic impacts.**

Credibility of the measures to enhance the career perspectives and employability of the researcher (including the contribution of their skills development):

Excellent

Suitability and quality of the planned dissemination and exploitation measures:

Excellent

Suitability and quality of the planned communication and public engagement activities:

Excellent

Strategy for intellectual property management and protection:

Not applicable

Expected scientific impact beyond the immediate scope and duration of the project:

Very good

Expected economical/ technological and/or societal impacts of the project beyond its immediate scope and duration:

Fair

Comments:

The proposal fails to properly address the magnitude, importance and credibility of its scientific impact. The proposal lacks a clear identification of the reasons for the claimed technological and societal impacts.

Criterion 3 - Implementation

Score: **4.80** (Threshold: 0 / 5.00 , Weight: 20.00%)

- **Quality and effectiveness of the work plan, assessment of risks and appropriateness of the effort assigned to work packages.**
- **Quality and capacity of the host institutions and participating organisations, including hosting arrangements.**

Quality and effectiveness of the work plan, including deliverables and milestones:

Good

Appropriateness of effort and timing assigned to each work package:

Excellent

Completeness of the mandatory Gantt Chart and its consistency with the overall work plan:

Excellent

Consideration of risks and adequacy of contingency measures:

Excellent

Quality of hosting arrangements, including team integration and support services:

Excellent

Quality and capacity of all participating organisations, including their infrastructure, logistics, and facilities:

Excellent

Comments:

The work plan is too ambitious for the duration of the project, in particular with respect to the number of deliverables. Efforts and timing for complex theoretical tasks are underestimated.

Scope of the application

Status: **Yes**

Comments (in case the proposal is out of scope)

Not provided

Exceptional funding

A third country participant/international organisation not listed in the General Annex to the Main Work Programme may exceptionally receive funding if their participation is essential for carrying out the project (for instance due to outstanding expertise, access to unique know-how, access to research infrastructure, access to particular geographical environments, possibility to involve key partners in emerging markets, access to data, etc.). (For more information, see the HE programme guide)

Please list the concerned applicants and requested grant amount and explain the reasons why.

Based on the information provided, the following participants should receive exceptional funding:

Not provided

Based on the information provided, the following participants should NOT receive exceptional funding:

Not provided

Use of human embryonic stem cells (hESC)

Status: **No**

If YES, please state whether the use of hESC is, or is not, in your opinion, necessary to achieve the scientific objectives of the proposal and the reasons why. Alternatively, please state if it cannot be assessed whether the use of hESC is necessary or not, because of a lack of information.

Not provided

Use of human embryos

Status: **No**

If YES, please explain how the human embryos will be used in the project.
Not provided

Activities excluded from funding

Status: **No**

If YES, please explain.
Not provided

Exclusive focus on civil applications

Status: **Yes**

If NO, please explain.
Not provided

Overall comments

Not provided

European Commission

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